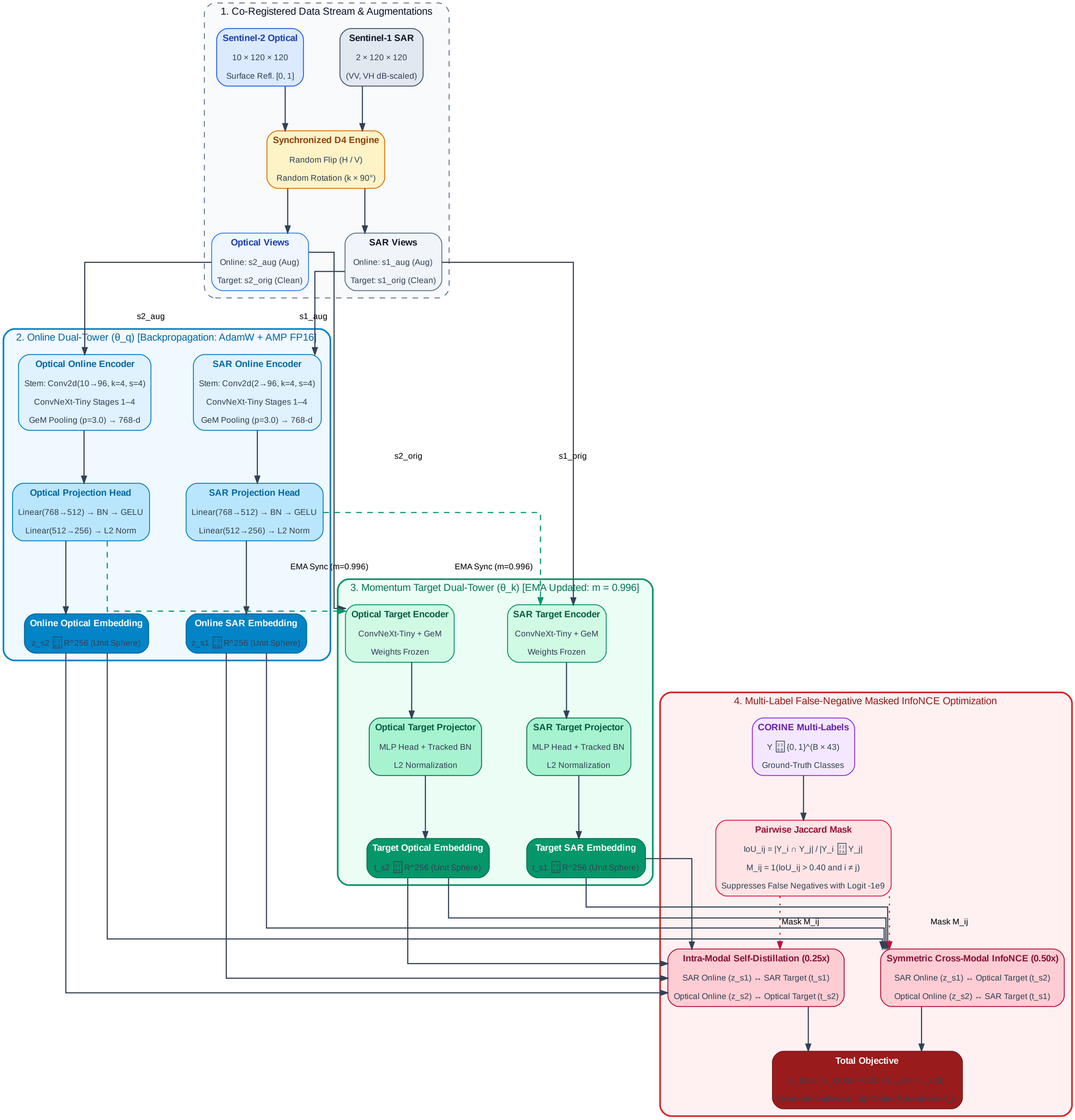
1. Co-Registered Data Stream & Augmentations
Sentinel-2 Optical
10 × 120 × 120
Surface Refl. [0, 1]
Sentinel-1 SAR
2 × 120 × 120
(VV, VH dB-scaled)
Synchronized D4 Engine
Random Flip (H / V)
Random Rotation (k × 90°)
Optical Views
Online: s2_aug (Aug)
Target: s2_orig (Clean)
SAR Views
Online: s1_aug (Aug)
Target: s1_orig (Clean)
s2_aug
s1_aug
2. Online Dual-Tower (θ_q) [Backpropagation: AdamW + AMP FP16]
Optical Online Encoder
Stem: Conv2d(10→96, k=4, s=4)
ConvNeXt-Tiny Stages 1–4
GeM Pooling (p=3.0) → 768-d
SAR Online Encoder
Stem: Conv2d(2→96, k=4, s=4)
ConvNeXt-Tiny Stages 1–4
GeM Pooling (p=3.0) → 768-d
Optical Projection Head
Linear(768→512) → BN → GELU
Linear(512→256) → L2 Norm
SAR Projection Head
Linear(768→512) → BN → GELU
Linear(512→256) → L2 Norm
Online Optical Embedding
z_s2 ∈ R^256 (Unit Sphere)
Online SAR Embedding
z_s1 ∈ R^256 (Unit Sphere)
s2_orig
s1_orig
EMA Sync (m=0.996)
EMA Sync (m=0.996)
3. Momentum Target Dual-Tower (θ_k) [EMA Updated: m = 0.996]
Optical Target Encoder
ConvNeXt-Tiny + GeM
Weights Frozen
SAR Target Encoder
ConvNeXt-Tiny + GeM
Weights Frozen
Optical Target Projector
MLP Head + Tracked BN
L2 Normalization
SAR Target Projector
MLP Head + Tracked BN
L2 Normalization
Target Optical Embedding
t_s2 ∈ R^256 (Unit Sphere)
Target SAR Embedding
t_s1 ∈ R^256 (Unit Sphere)
4. Multi-Label False-Negative Masked InfoNCE Optimization
CORINE Multi-Labels
Y ∈ {0, 1}^(B × 43)
Ground-Truth Classes
Pairwise Jaccard Mask
IoU_ij = |Y_i ∩ Y_j| / |Y_i ∪ Y_j|
M_ij = 1(IoU_ij > 0.40 and i ≠ j)
Suppresses False Negatives with Logit -1e9
Mask M_ij
Mask M_ij
Intra-Modal Self-Distillation (0.25x)
SAR Online (z_s1) ↔ SAR Target (t_s1)
Optical Online (z_s2) ↔ Optical Target (t_s2)
Symmetric Cross-Modal InfoNCE (0.50x)
SAR Online (z_s1) ↔ Optical Target (t_s2)
Optical Online (z_s2) ↔ SAR Target (t_s1)
Total Objective
L_total = L_cross + 0.25 × (L_sar + L_opt)
Gradients backward into Online Parameters θ_q